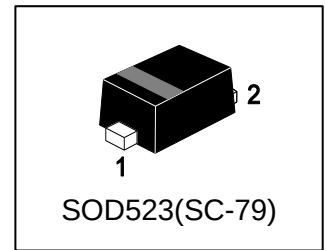


LESD5D5.0CT1G

Transient Voltage Suppressors for ESD Protection

1. FEATURES

- We declare that the material of product compliance with RoHS requirements and Halogen Free.
- ESD rating of Class 3 per Human Body Model
- Small body outline dimensions
- Low leakage
- Response time is typically < 1.0 ns
- IEC61000-4-2 level 4 ESD protection
- IEC61000-4-4 Level 4 EFT protection



2. APPLICATIONS

- Thunderbolt interface
- USB3.1 and USB3.0 interfaces
- USB Type-C interface
- DisplayPort interface
- Hand held portable applications
- Mobile Phones
- Consumer electronics

3. DEVICE MARKING AND ORDERING INFORMATION

Device	Marking	Shipping
LESD5D5.0CT1G	C	5000/Tape&Reel

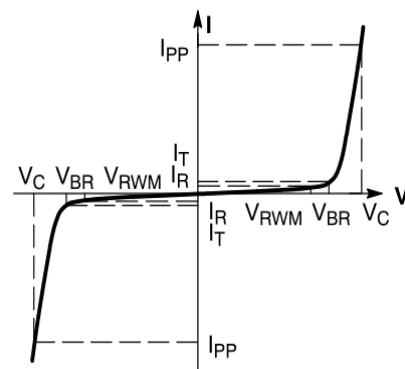
4. MAXIMUM RATINGS(Ta = 25°C)

Parameter	Symbol	Limits	Unit
IEC 61000-4-2 (ESD) Contact		± 8	KV
Air		± 15	
Peak Pulse Power @8/20 μs(Note 1)	PPP	174	W
Peak pulse current @8/20 μs(Note 1)	IPP	9.4	A
Maximum Junction Temperature	Tj	150	°C
Operating Temperature Range	Top	-40 ~ +125	°C
Lead Solder Temperature – Maximum (10 Second Duration)	TL	260	°C
Storage temperature	Tstg	-55 ~ +155	°C

Note 1.Surge current waveform per Figure 1 according to IEC 61000-4-5.

5. ELECTRICAL CHARACTERISTICS (Ta= 25°C)

Symbol	Parameter
IPP	Maximum Reverse Peak Pulse Current
VC	Clamping Voltage @ IPP
VRWM	Working Peak Reverse Voltage
IR	Maximum Reverse Leakage Current @ VRWM
VBR	Breakdown Voltage @ IT
IT	Test Current
Ppk	Peak Power Dissipation
C	Capacitance @ VR = 0 and f = 1.0 MHz


6. ELECTRICAL CHARACTERISTICS (Ta= 25°C)

Characteristic	Symbol	Min.	Typ.	Max.	Unit
Reverse stand-off voltage	VRWM			5.0	V
Reverse leakage current (VRWM = 5.0 V)	IRM			1	μA
Breakdown voltage (IT = 1 mA)	VBR	5.6		7.8	V
Clamping Voltage(Note 1) (IPP = 5 A)(8 x 20μs pulse) (IPP = 9.4 A)(8 x 20μs pulse)	VC		11.6	18.6	V
Junction Capacitance (VR = 0V, f = 1MHz)	CJ		15		pF

Note 1.Surge current waveform per Figure 1 according to IEC 61000-4-5.

7. ELECTRICAL CHARACTERISTICS CURVES

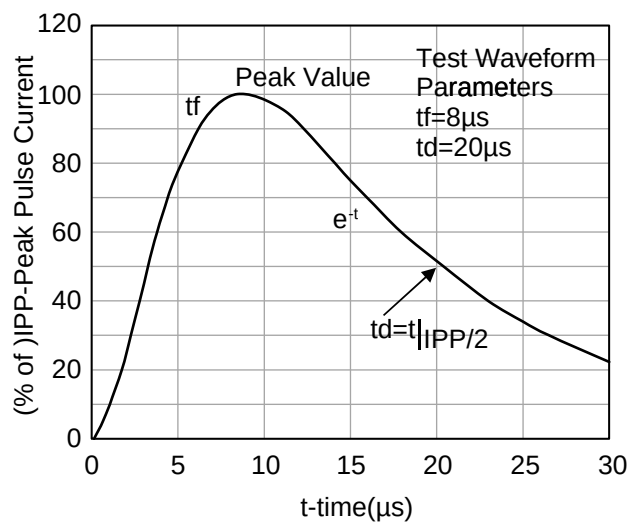


Figure 1. Pulse Waveform according to IEC 61000-4-5

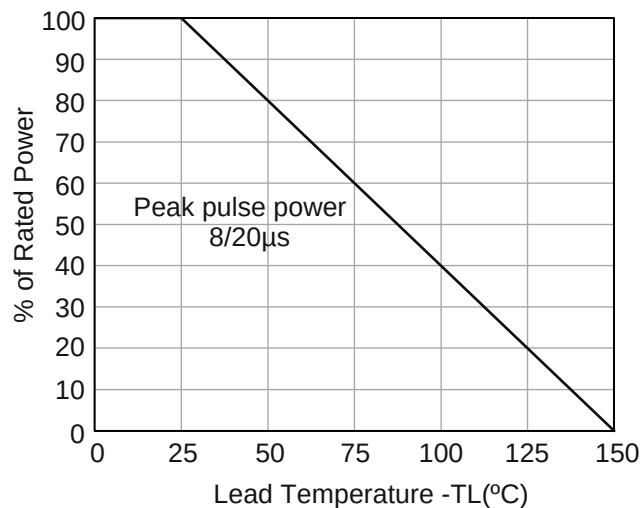
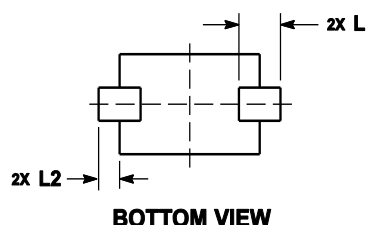
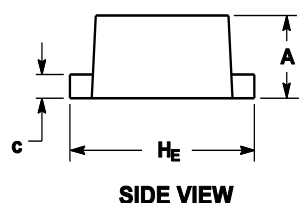
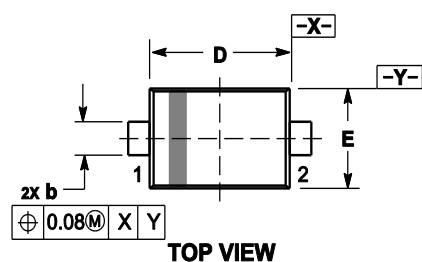


Figure 2. Power Derating Curve

8.OUTLINE AND DIMENSIONS

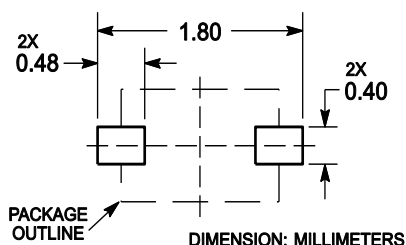


Notes:

1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M, 1982.
2. CONTROLLING DIMENSION: MILLIMETERS.
3. MAXIMUM LEAD THICKNESS INCLUDES LEAD FINISH. MINIMUM LEAD THICKNESS IS THE MINIMUM THICKNESS OF BASE MATERIAL.
4. DIMENSIONS D AND E DO NOT INCLUDE MOLD FLASH, PROTRUSIONS OR GATE BURRS.

DIM	MILLIMETERS			INCHES		
	MIN	NOM	MAX	MIN	NOM	MAX
A	0.50	0.60	0.70	0.020	0.024	0.028
b	0.25	0.30	0.35	0.010	0.012	0.014
c	0.07	0.14	0.20	0.003	0.006	0.008
D	1.10	1.20	1.30	0.043	0.047	0.051
E	0.70	0.80	0.90	0.028	0.031	0.035
H _E	1.50	1.60	1.70	0.059	0.063	0.067
L	0.30 REF			0.012 REF		
L ₂	0.15	0.20	0.25	0.006	0.008	0.010

9.SOLDERING FOOTPRINT



DISCLAIMER

- Curve guarantee in the specification. The curve of test items with electric parameter is used as quality guarantee. The curve of test items without electric parameter is used as reference only.
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- All information contained in this document is current as of the issuing date and subject to change without any prior notice. Before purchasing or using LRC's Products, please confirm the latest information with a LRC sales representative.